Universität St.Gallen

Natural Language Processing with Large Language Models

Exercises Spring Term 2026

Prof. Dr. Siegfried Handschuh
Teaching Assistants: Götz-Henrik Wiegand & Lorena Raichle

DS-NLP Lab

From insight to impact.

Teaching Assistants

Contact us via **Teams message!**

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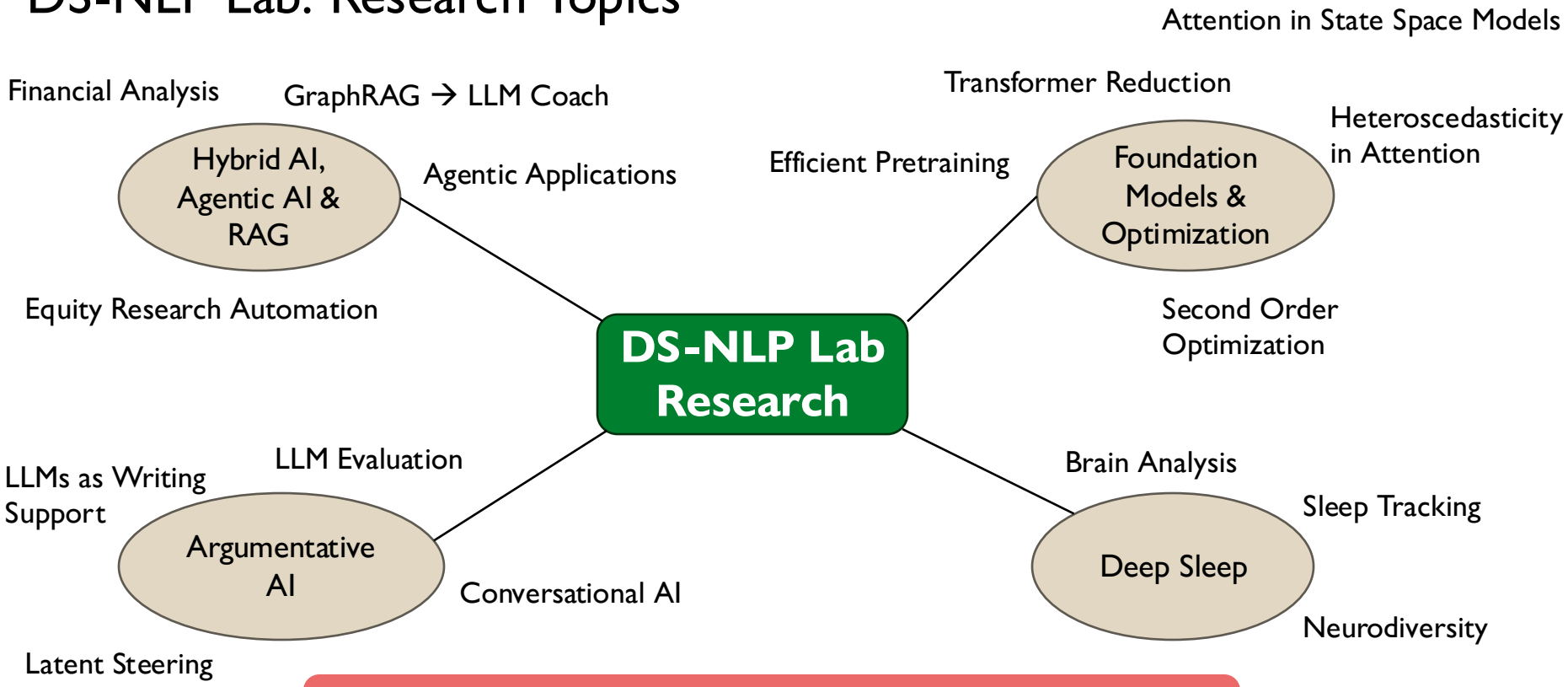
1. DS-NLP Lab: Research Topics

DS-NLP Lab: Research Topics



Interested in a **Master's project (IMP)** or **Thesis topic**?
Get in touch with us!

Quantitative Data Science Research Topics

A Convexity-dependent Two-Phase Training Algorithm for Deep Neural Networks

Tomas Hrycej[1†], Bernhard Bermeitinger[2†], Massimo Pavone[1‡], Götz-Henrik Wiegand[1†], and Siegfried Handschuh[1†]

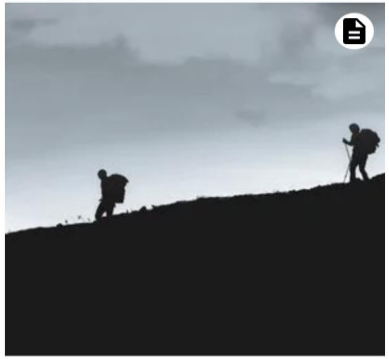
[1] *Institute of Computer Science, University of St.Gallen (HSG), St.Gallen, Switzerland*

[2] *Institute of Computer Science in Vorarlberg, University of St.Gallen (HSG), Dornbirn, Austria*

[†] *firstname.lastname@unisg.ch*, [‡] *firstname.lastname@student.unisg.ch*

Integrating the Attention Mechanism into State Space Models

Reducing the Transformer Architecture to a Minimum



Research - 18.11.2025 - 10:00

HSG researchers develop more efficient method for training AI

Interested in a **Master's project (IMP)** or **Thesis topic**?
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2. Why Understanding LLM Training matters now

Why Understanding LLM Training matters now

"Right now, people talk about being an AI company. There was a time after the iPhone App Store launch when people talked about being a mobile company. But no software company says they're a mobile company now because it'd be unthinkable not to have a mobile app. And it'll be unthinkable not to have intelligence integrated into every product and service. It'll just be an expected, obvious thing."

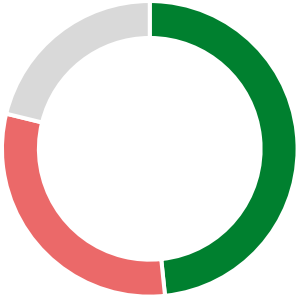
— Sam Altman, co-founder and CEO, OpenAI

- **LLMs** are becoming **infrastructure**
- **Understanding training** beats using APIs
- **Efficiency** & **constraints** define real-world AI → Cost, energy, latency, deployment matter
- **This skill compounds** → Research, startups, industry, … (all depend on it)

Foundation Models Are Becoming Core Infrastructure
From Single-Task Models to General Purpose Technology

Global **LLM market** growth: from **~$9B** (2025) to **~$68B** by 2032 (34% compound annual growth rate)

Global LLM Market Revenue Share by Segment (2025)[1]



- 48 % Foundation Model Segment
- 30 % Conversational Agents & Virtual Assistants
- Other LLM Revenue

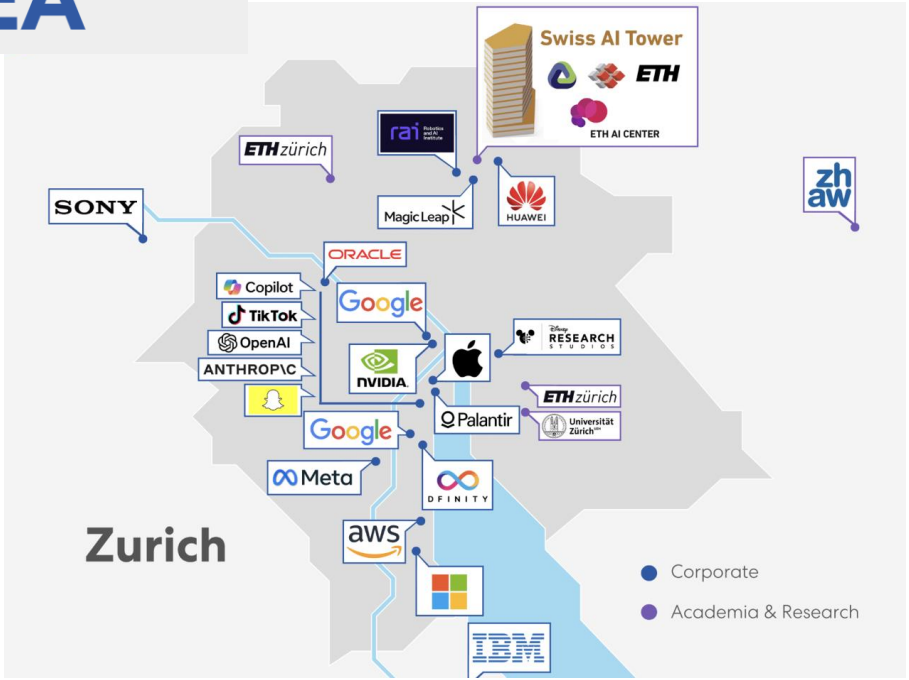
Largest share of **LLM value** is created at **foundation model layer**

- By 2023, **industry** accounted for ~72% of all new foundation models
- Rise of Efficient/Small Language Models

[1] Coherent Market Insights: https://www.coherentmarketinsights.com/industry-reports/large-language-model-market

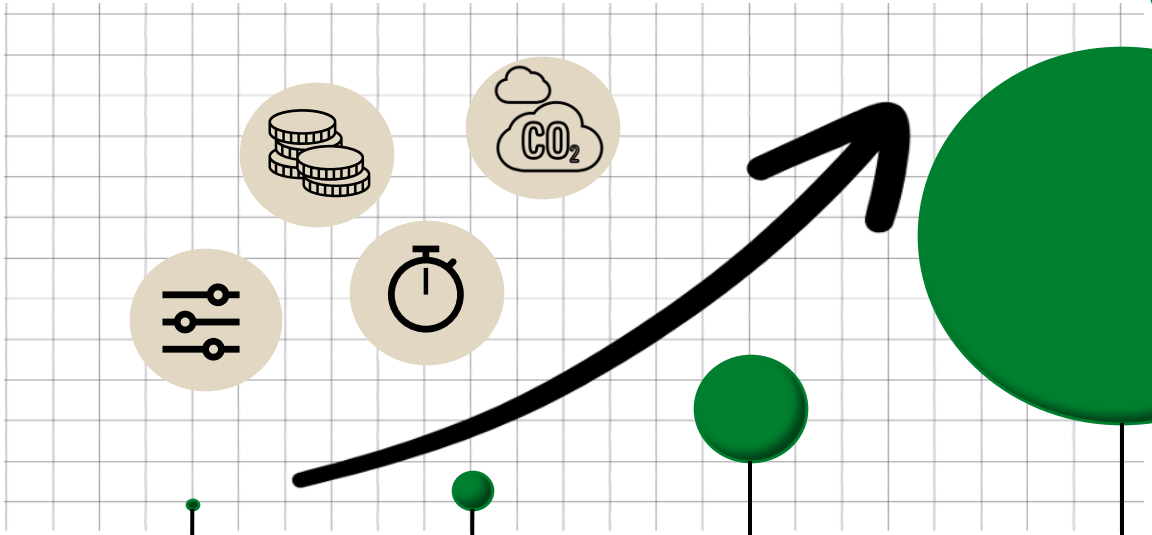
GREATER ZURICH AREA

More than 70% of the organizations featured are Swiss-made, underscoring the region's strength in home-grown innovation and deep technical expertise. At the same time, global players such as Anthropic, NVIDIA, OpenAI, and Pinterest have chosen Zurich because of this very ecosystem. They rely on its top talent, its research excellence, and its uniquely collaborative environment, often working side by side with local startups, university labs, and established tech companies.



https://www.greaterzuricharea.com/de/die-zurich-ai-map-unternehmen-start-ups-und-forschungsinstitute-der-region-zuerich

Why **efficient** Language Model Training matters



"Attention is all you need" [1]

GPT-1 **117 M**

GPT-2 **1,5 B**

GPT-3 **175 B**

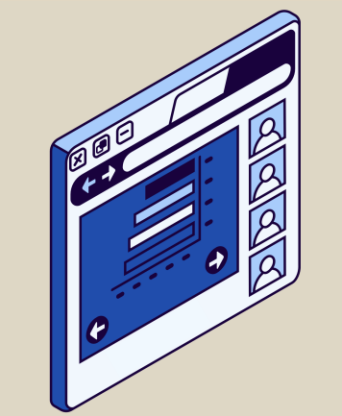
GPT-4 **1.8 T**

GPT-5 ?

2017 — 2025

[1]Vaswani, Ashish, Noam Shazeer, Niki Parmar, u. a. „Attention is all you need". *Proceedings of the 31st International Conference on Neural Information Processing Systems* (Red Hook, NY, USA), NIPS'17, 4. Dezember 2017, 6000–6010.

3. *PikoGPT* Challenge Introduction



From Zero to Demo: The *PikoGPT* Challenge

Building a tiny Language Model in 12 Weeks from Scratch!*

What you build	*In Teams of 4 students, build a 10–40M param model trained end-to-end*
Why it matters	*Cost · Energy · Efficiency · Sustainability*
What makes it hard	*No wrapper. No APIs. No pre-trained checkpoints.* *Realistic compute & data constraints.*
How you win	*Leaderboard Performance + Engineering Quality + Poster & Demo Session*

Training an LLM Is an Investment Decision
Scaling, Compute and Real-World Constraints

Compute cost scales with
- model size & data
- hyperparameter search

How Large Labs Train LLMs:
- Explore architectures & hyperparameters at small scale
- Use **scaling laws**[1] to extrapolate relative performance trends
- Run only a few large, expensive training jobs

Small models → Scaling Laws → Large-model assumptions → GPU investment

Compute is capital – You get **2x 24 h GPU resources** as investor funding!

! Convince us to invest with your results and training strategy.

[1] Kaplan, Jared, Sam McCandlish, Tom Henighan, u. a. „Scaling Laws for Neural Language Models". arXiv:2001.08361. Preprint, arXiv, 23. Januar 2020. https://doi.org/10.48550/arXiv.2001.08361.

Challenge Fact Sheet: Technical Constraints

Hard Constraints & Requirements

- **Pretraining Data:** Provided OpenWebText subset.
- **Tokenizer:** GPT-2 tokenizer

```
from transformers import GPT2TokenizerFast
tokenizer = transformers.GPT2TokenizerFast.from_pretrained("gpt2")
```

- **Architecture:** You are encuraged to design your own Architecture within the following constrains!
  - Decoder-Only - No MoE
  - Context size: 1024
  - Max Decoder-Layers: 24
  - Model Size: Max. 40M parameters.
- **Compute Budget:** 2x 24h on 8xV100 (or similar)
- **Tech Stack:** Python & PyTorch. High-level frameworks for the training are forbidden for the main parts of the training and repository. **If you are unsure, ask the TAs.**
- **Forbidden:** Pretraining on extra data, using external checkpoints, knowledge distillation, etc.

Deadlines & Submission

	Date	Room	Exercise Content	Milestones
1	17.02	Säntis - 1.OG	Motivation & Challenge Intro	**Startup Registration**
2	24.02	Säntis - 1.OG		
3	03.03	Hilti Inn. R1 - 1.OG		
4	10.03	Winterg. - 1.OG		
5	17.03	Hilti Inn. R2 - 1.OG		
6	24.03	Rosenb. Rotm. - EG		
			— Semester Break —	
			— Semester Break —	
	13.04			**Slide submission via Canvas**
7	14.04	Vadian Gallus - EG		**Midterm Presentation**
8	21.04	Vadian Gallus - EG		
9	28.04	Bodensee - 1.OG		
10	05.05	C, 61 61-070		
	11.05			**Poster submission via Canvas**
11	12.05	Hilti Inn. R2 - 1.OG		
12	19.05	Hilti Inn. R1 - 1.OG	**Pseudo-Conference**	**Demo & Poster**
	02.06		**Project Submission**	**Code + Report Submission**

4. From Zero to Demo:

How to Design, Train, and Evaluate Your Own LLM

Lecture & Exercises

From Theory to a Trained Model

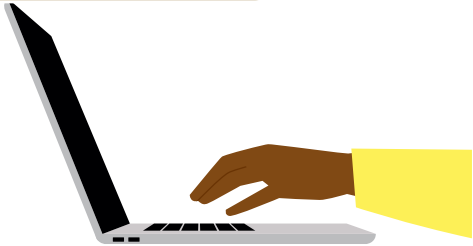
⟶ The **lecture theory** is needed for the **practical challenge**!

Why?

Lecture Focus:

- Master the Model Architecture
- Navigate critical design trade-offs

How?



Exercises Focus:

- Apply lecture concepts in **code**
- Experiment with: Architectures, Hyperparameters, Data strategies
- Prepare your model for final GPU runs

Exercises: Time to work on your Foundation Model

~15 min *Theory in Code*

- Code input on this week's lecture topic
- Tips & Tricks

Co-Working & Mentoring Time

Sync with your co-founders and build your product.

- Ask questions & get individual support
- Work **onsite** with your group

Roadmap: Building an LLM from Scratch

Chat Demo

1 **Data Preprocessing**

- data quality
- data limits
- filtering choices

2 **Pretraining**

- Architecture & hyperparams
- Training stability
- Compute budget

3 **Post-training**

- Instruction tuning
- Alignment trade-offs
- SFT and DPO

4 **Evaluation**

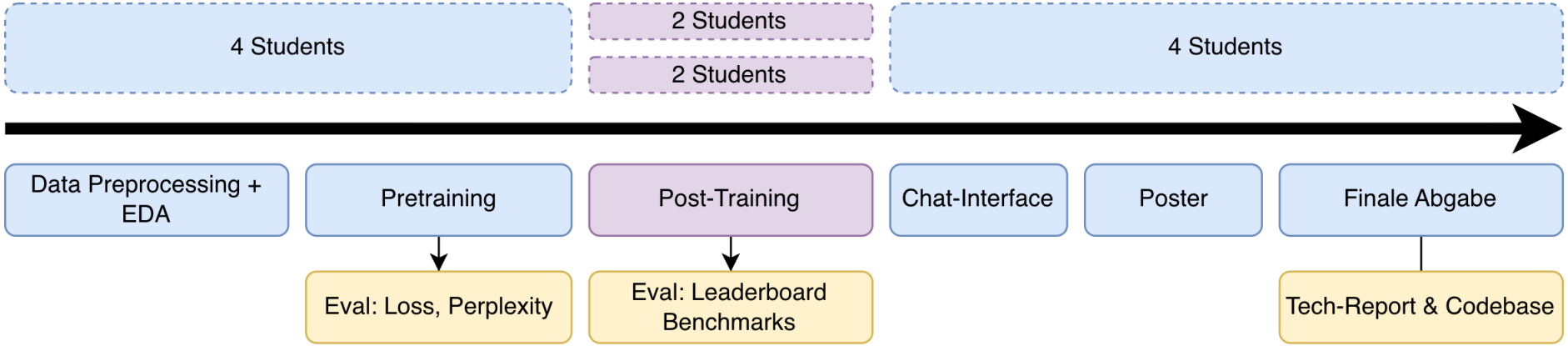
- Metric reliability
- Generalization testing
- Leaderboard

Test model instantiation

Inference Stage for rapid Model-Testing

Iteration & Refinement

Roadmap: Group Phases



0 Planning: Group Finding and Milestone Planning (**Today!**)

- Find a group of **4 students** to start your startup!
- Chose a convincing **startup name** that fits to the challenge
- **Plan milestones** (backward planning)
  - Start at the Deadlines and work your way back
  - Define rough goals for every Month/Week
  - It is okay if the plan shifts later
  - Think about roles/distribution of work in your startup
- Convince us!
  - Tell us your group members and startup name
  - Propose your milestones to TAs
  - Get feedback and approval from TAs
- Start implementing!

https://github.com/unisg-ics-dsnlp/PikoGPT_Template

Repository Template for you to start!

1 Data Preprocessing + EDA: From Text to Training Data

- All groups use the *NLP26_OpenWebText* split
  - Excludes a Test-Split for testing later
  - You are free to complement with other datasets (except data from the Wikitext103 and NLP26_OpenWebText test splits)
- Start with **"Exploratory Data Analysis (EDA)"** in a separate Notebook
  - Understand the Dataset
- Implement a **Data Preprocessing Stage** that:
  - Downloads the dataset
  - Cleans up the dataset
  - Make sure your data does not contain the test splits.
  - Prepares the dataset for pretraining (tokenization etc.)
- You might revisit this step for refinement later

Wikitext103 official test split from HF:
https://huggingface.co/datasets/Salesforce/wikitext/viewer/wikitext-103-raw-v1/test

NLP26 OpenWebText evaluation test set:
https://drive.switch.ch/index.php/s/6TLGQFEIkAPJ72K

2 Pretraining: Teaching the Model Language

Architecture

- No MoE, decoder-only!
- 20M – 50M params
- Max. context length: 1024
- Max. decoder blocks: 24
- All use the GPT-2 Tokenizer

- Test and improve variants (e.g., RoPE, Flash Attention) on small scales first.

- **Hyperparameter Tuning**: Systematically explore: learning rates, batch sizes…

- More infos in Fact-Sheet!

Every decision must be justified in your **Tech Report**.

Forbidden!

× NO Training on Test-Data: Train only on the provided OpenWebText split and other data that is not included in test

× NO **External Checkpoints**: You must initialize from random weights.

× NO **Distillation**: Training on outputs from other LLMs.

Ask TAs if anything is unclear or if you have questions!

2 Pretraining: Teaching the Model Language

Goal: Train the model to have a **complex understanding of the english language!**

- Most **GPU-Cost intense** step
- One run can take 24h or more. Test with small experiments on your own hardware /Google Colab
- Use *seeds* for reproducibility
- Use *Scaling Laws* to estimate the perfect Hyperparameters, Dataset Size, Model Size/Architecture, …
- Make sure:
  - to have training evals, logging, checkpointing, learning rate scheduling etc.
  - the training and inference can run on CUDA GPUs (V100 and H100) and on MPS (Mac Arm Architecture)
    - V100 don‘t support Flash Attention!
    - Mac is used for Leaderboard benchmarking
  - One training script that starts the training
    - We give you **2 x 24h GPU resources** in a docker container with 8xV100 or similar.

- Come to us, show us your results, convince us to invest the GPU-Time!

3 Post-Training

Goal: Train the model on the desired behavior!

- You can do *SFT, DPO, RLHF* etc. (More about that in the lecture!)
- Try to **maximize the Benchmarks** on the Leaderboard
- Try to avoid *catastrophic forgetting* → Don‘t overlearn on one task!
- The model should be capable of various tasks: Multiple Choice, Light Reasoning etc.
  - Coding is excluded from the benchmarks
  - We only focus on english text

4 Benchmarks & Leaderboard

There are more hidden benchmarks!

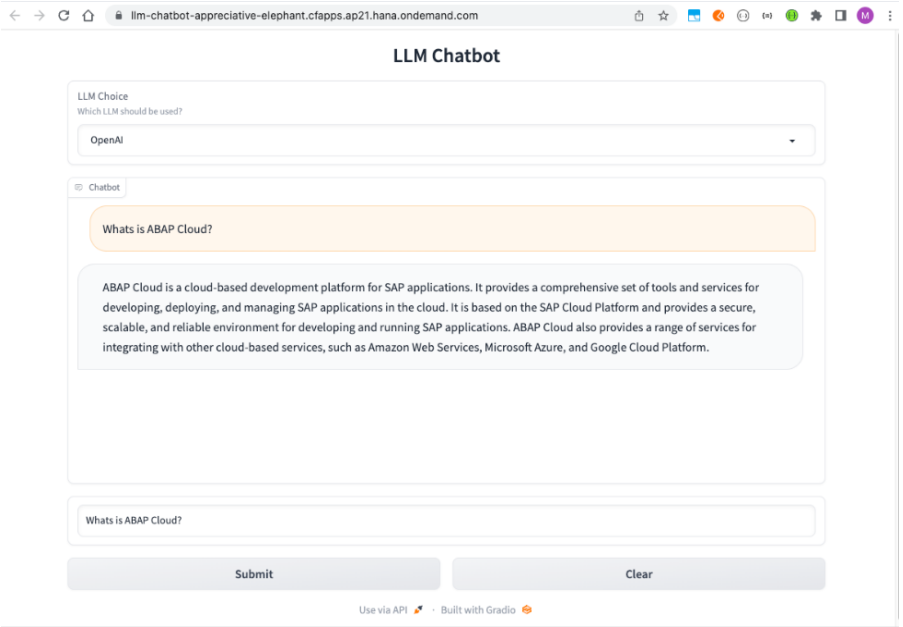
1. Contextual Understanding
LAMBADA (OpenAI, 2016)
https://arxiv.org/abs/1606.06031
Task: Next-Word Prediction in broad context.

2. Commonsense Reasoning (Multiple Choice)
HellaSwag (Zellers et al., 2019)
https://huggingface.co/datasets/Rowan/hellaswag
Winogrande (AI2, 2019)
https://huggingface.co/datasets/allenai/winogrande

3. Knowledge & Logic
OpenBookQA (AllenAI, 2018)
https://huggingface.co/datasets/allenai/openbookqa/viewer/additional/train?row=0

Small Chat Interface

- Goal: Simple Demo Interface
- Just as showcase
- Frameworks are allowed!
- Part of Pseudo-Conference (Demo-Session)
- Keep it Simple!
- See factsheet for more Infos.



Final GPU Runs

The Hardware

- **2x 24h on 8x V100 GPUs (**Or something similar)
  - V100 dont have Flash Attention!
- Engineere on your own devices or Google Colab
- Come to TAs with concept for training
- After Approval you get 24h on the GPUs
- **Session gets deleted after 24h** → Make sure to save all checkpoints etc.
- You decide how you invest the 24h for each run!

Tip: Focus on Pretraining runs on the GPUs!

- Post-training can go on google colab or on local hardware

Poster Conference & Demo Session

Showing the Model in Action

There will be rewards!

Poster

- A professional, academic-style poster
- summarizing your results and decisions
- Summarize the key findings
- Make it visually appealing
- Include Plots, Tables, Visualisations etc.

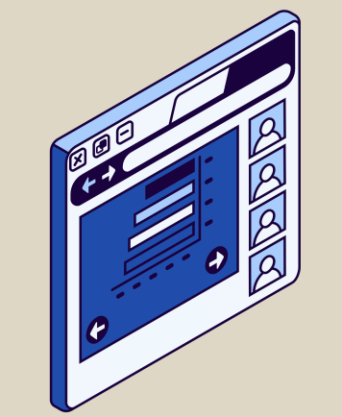
- We provide a optional template!

Demo

- Run your chat interface in class
- We prepare some prompts that all LLMs need to answer
- Be able to switch out different checkpoints (see group phases)
- Testing Pre-Training checkpoint and Post-Training checkpoints
- Needs to run on a Macbook M3 (MPS)

Showcase your results (Demo and Poster) and **discuss** with others!

5. Grading & Deadlines



From Zero to Demo: The *PikoGPT* Challenge

Training a tiny Language Model in 12 Weeks from Scratch!

Course Deliverables:

13. April:	***Slide Submissions***
14. April:	***Intermediate Presentation***
11. Mai:	***Poster PDF Submission for Printing***
02. June:	***Final Code, Documentation and Report Submission***

Grading Criteria

- **10%** Active lecture and excercise participation
- **65%** Project
  - Repository: Code, Documentation, Readme
  - Benchmarks: Leaderboard Results, Pre-Training evals (Loss, Perplexity)
  - Tech Report
  - Project Work and Participation
- **5%** Midterm Presentation
- **20%** Pseudo Conference
  - Poster
  - Demo Session
  - Conference participation (Poster Discussions etc.)

6. Deliverables

Midterm Presentation

- Present your current status
- Max. 10 Slides
- **10 Minutes** per startup
- Pitch style to inform the "Investors" about status quo
- Hand in your slides the day bevor the presentation via canvas (Deadline: 13.04.2026)

Should Haves:

- **Architecture** and Project Decisions & Reasoning
- **Benchmarks** from first Pre-training
- Overview on Project Status
- Potential Failures
- **Examples** (LLM inference tests etc.)

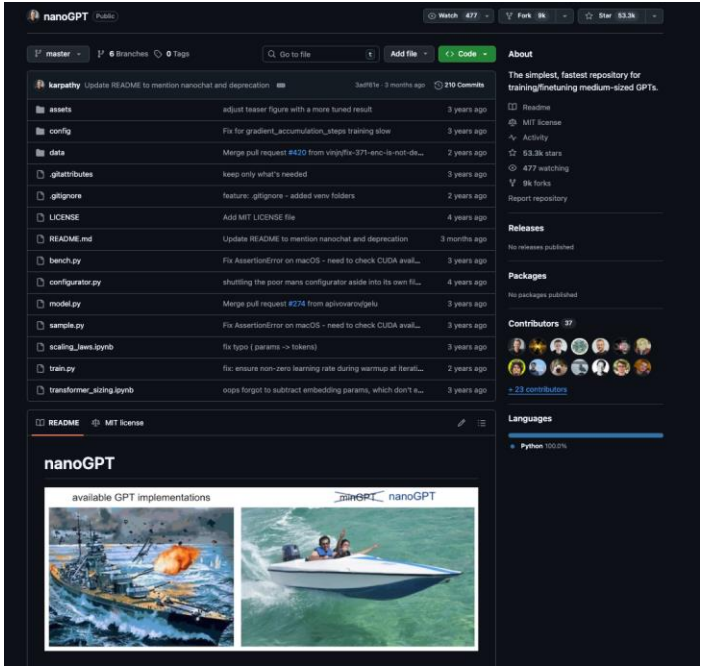
Could Haves:

- Post-Training Ideas
- Unique ideas that set you appart
- And more...

Repository

- Must contain:
  - Clean Code
  - Documentation (Readme, Docstrings etc.)

- Important:
  - Reproducibility (Use UV as packagemanager, Clear Structure in Stages, Deterministic code with Seeds etc.)
  - Cleaned Codebase
  - Short Project description
  - Good documentation (Visuals, instructions to run etc.)



The repository should be easy **understandable** and **tidy**!

Pseudo Conference

- Structured in two parts:
  - Demo Session
  - Poster Session
- Conference closing with rewards!
- Other members of our lab will join and discuss your results.
- More Infos on the pseudo conference after the break!

A conference is the best way to share and discuss knowledge!

Universität St.Gallen

ICML International Conference On Machine Learning

SDS2025 IEEE SWISS CONFERENCE ON DATA SCIENCE

ICLR International Conference On Learning Representations

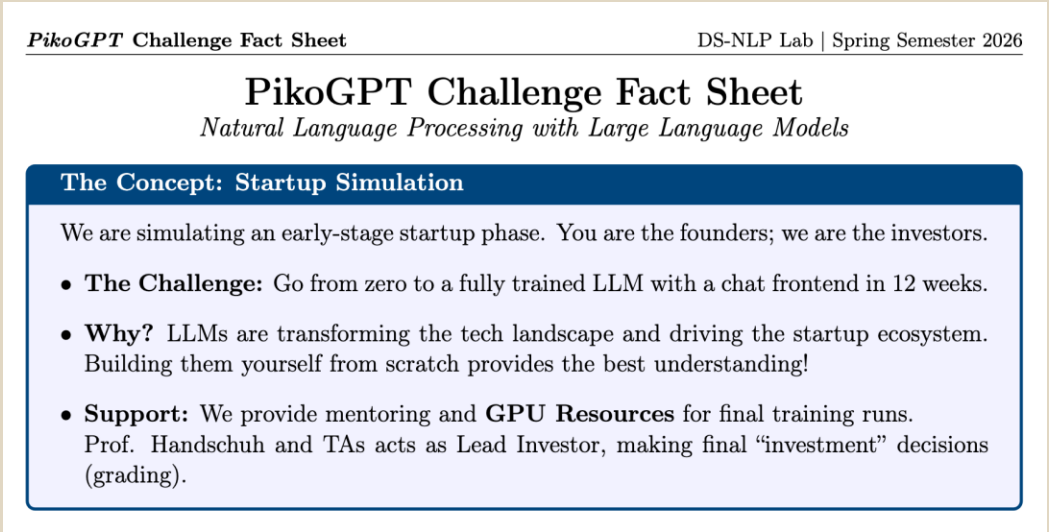
ASSOCIATION FOR COMPUTATIONAL LINGUISTICS

7. Next Steps

Co-Working & Mentoring Time

Questions? Help?

Check out the **Challenge Fact Sheet** on Canvas!



***PikoGPT* Challenge Fact Sheet** DS-NLP Lab | Spring Semester 2026

PikoGPT Challenge Fact Sheet

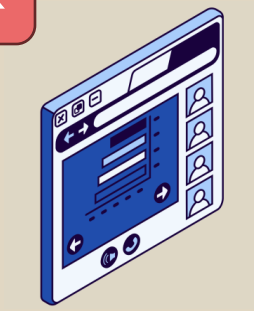
Natural Language Processing with Large Language Models

The Concept: Startup Simulation

We are simulating an early-stage startup phase. You are the founders; we are the investors.

- **The Challenge:** Go from zero to a fully trained LLM with a chat frontend in 12 weeks.
- **Why?** LLMs are transforming the tech landscape and driving the startup ecosystem. Building them yourself from scratch provides the best understanding!
- **Support:** We provide mentoring and **GPU Resources** for final training runs. Prof. Handschuh and TAs acts as Lead Investor, making final "investment" decisions (grading).

Task



Next Steps

- ✓ Form **groups** of 4 students (if you can't find a group, we will assist)
- ✓ Choose a **Startup Name**
- ✓ Plan **milestones** (backwards planning)
- ✓ **Submit** your group members, startup name and milestones to us **today!**
- ✓ Start with Data Preprocessing Concept / EDA